THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 3: Matrices (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	Let $A = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 4 & -1 \\ -3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -2 \\ -5 \\ -7 \end{bmatrix}$, $C = [9 \quad 8 \quad 7]$. Which of the following is defined? (A) Only AB (B) Only AC (C) Only BA (D) All AB, AC and BA	1
2.	If $\begin{bmatrix} x+y & 3y \\ 3x & x+3 \end{bmatrix} = \begin{bmatrix} 9 & 4x+y \\ x+6 & y \end{bmatrix}$, then $(x-y)$ is equal to: (A) -7 (B) -3 (C) 3 (D) 7	1
3.	If $A = [a_{ij}]$ is a skew-symmetric matrix of order n, then a_{ii} is equal to: (A) 1 (B) 0 (C) -1 (D) n	1
4.	Let $A = [a_{ij}]$ be a 2×2 matrix whose elements are given by $a_{ij} = \frac{(i+2j)^2}{3}$. Then A' is: (A) $\begin{bmatrix} 12 & \frac{25}{3} \\ \frac{16}{3} & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 12 & \frac{16}{3} \\ \frac{25}{3} & 3 \end{bmatrix}$ (C) $\begin{bmatrix} 3 & \frac{25}{3} \\ \frac{16}{3} & 12 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & \frac{16}{3} \\ \frac{25}{3} & 12 \end{bmatrix}$	1

Q No.	Question	Marks
5.	If A is a square matrix such that $A^2 = A$, then $(I + A)^3 - 7A$ is equal to: (A) A (B) $I - A$ (C) I (D) $3A$	1
6.	**Assertion (A):** If A and B are two square matrices such that AB and BA are defined, then it is not necessary that $AB = BA$. **Reason (R):** The product of two diagonal matrices of the same order is commutative. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Find x and y if $2\begin{bmatrix}1 & 3\\0 & x\end{bmatrix} + \begin{bmatrix}y & 0\\1 & 2\end{bmatrix} = \begin{bmatrix}5 & 6\\1 & 8\end{bmatrix}$.	2
	LEVEL 2 **Q 8 – Q 14**	**18**
8.	Construct a 3×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \frac{1}{2}\lvert i - 3j\rvert$.	2
9.	If $A = \begin{bmatrix}1 & 2\\2 & 1\end{bmatrix}$, show that $A^2 - 2A - 3I = O$. Hence find A^{-1}.	3
10.	Express $A = \begin{bmatrix}3 & 5\\1 & -1\end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.	3
11.	If $A = \begin{bmatrix}8 & 0\\4 & -2\\3 & 6\end{bmatrix}$ and $B = \begin{bmatrix}2 & -2\\4 & 2\\-5 & 1\end{bmatrix}$, find the matrix X such that $2A + 3X = 5B$.	2
12.	Find x if $\begin{bmatrix}x & -5 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 2\\0 & 2 & 1\\2 & 0 & 3\end{bmatrix}\begin{bmatrix}x\\4\\1\end{bmatrix} = O$.	3
13.	If $A = \begin{bmatrix}\cos\alpha & -\sin\alpha\\\sin\alpha & \cos\alpha\end{bmatrix}$, find $\alpha\left(0 < \alpha < \frac{\pi}{2}\right)$ such that $A + A' = I$.	2
14.	If $A = \begin{bmatrix}1\\-4\\3\end{bmatrix}$ and $B = [-1 \quad 2 \quad 1]$, verify that $(AB)' = B'A'$.	3
	LEVEL 3 **Q 15 – Q 20**	**23**
15.	If $A = \begin{bmatrix}2 & 0 & 1\\2 & 1 & 3\\1 & -1 & 0\end{bmatrix}$, find $A^2 - 5A + 4I$ and hence find a matrix X such that $A^2 - 5A + 4I + X = O$.	3
16.	A trust fund has ₹30,000 to be invested in two types of bonds paying 5% and 7% interest per annum. Using matrix multiplication, determine how to divide ₹30,000 among the two bonds so that the trust fund earns an annual interest of ₹1,800.	3

Q No.	Question	Marks
17.	If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, show that $A^2 = 3A$ and hence find A^3.	3
18.	If $A = \begin{bmatrix} 0 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 0 \end{bmatrix}$ and I is the identity matrix of order 2, show that $I + A = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$.	5
19.	If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}$, show that $A^3 - 23A - 40I = O$.	5
20.	**Case Study:** Three schools P, Q and R organised a fair to raise funds by selling hand-made fans, mats and plates made from recycled material at ₹25, ₹100 and ₹50 each respectively. School P sold 40 fans, 50 mats and 20 plates; School Q sold 25 fans, 40 mats and 30 plates; School R sold 35 fans, 50 mats and 40 plates. **(i)** Write the sales of the three schools as a 3×3 matrix and the prices as a column matrix. **[1]** **(ii)** Find the funds collected by School P. **[1]** **(iii)** Using matrix multiplication, find the funds collected by each school and the total funds collected. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 3: Matrices

Q No.	Answer / Key Steps
	LEVEL 1
1.	(A) $A_{3\times3}B_{3\times1}$ is defined; AC and BA are not.
2.	(B) $x = 3, y = 6$, so $x - y = -3$
3.	(B) 0
4.	(D) $A = \begin{bmatrix} 3 & \frac{25}{3} \\ \frac{16}{3} & 12 \end{bmatrix}$, so $A' = \begin{bmatrix} 3 & \frac{16}{3} \\ \frac{25}{3} & 12 \end{bmatrix}$
5.	(C) $(I + A)^3 = I + 3A + 3A^2 + A^3 = I + 7A$, so the result is I
6.	(B) Both true, but R is not the correct explanation of A.
7.	$y = 3, x = 3$
	LEVEL 2
8.	$A = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$
9.	$A^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$, $A^2 - 2A = 3I$. Multiplying $A^2 - 2A - 3I = O$ by A^{-1}: $A^{-1} = \frac{1}{3}(A - 2I) = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$.
10.	$P = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix}$ (symmetric), $Q = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ (skew-symmetric), $A = P + Q$.
11.	$X = \frac{1}{3}(5B - 2A) = \begin{bmatrix} -2 & -\frac{10}{3} \\ 4 & \frac{14}{3} \\ -\frac{31}{3} & -\frac{7}{3} \end{bmatrix}$
12.	Product $= x^2 - 48 = 0 \Rightarrow x = \pm 4\sqrt{3}$
13.	$2\cos\alpha = 1 \Rightarrow \alpha = \frac{\pi}{3}$
14.	$AB = \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix}$; $(AB)' = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix} = B'A'$.
	LEVEL 3
15.	$A^2 - 5A + 4I = \begin{bmatrix} -1 & -1 & -3 \\ -1 & -3 & -10 \\ -5 & 4 & 2 \end{bmatrix}$; $X = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{bmatrix}$
16.	$[x \quad 30000 - x]\begin{bmatrix} 0.05 \\ 0.07 \end{bmatrix} = 1800 \Rightarrow 2100 - 0.02x = 1800 \Rightarrow x = 15000$. ₹15,000 in each bond.
17.	$A^2 = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} = 3A$; $A^3 = A \cdot A^2 = 3A^2 = 9A$.
18.	Let $t = \tan\frac{\alpha}{2}$; use $\cos\alpha = \frac{1-t^2}{1+t^2}$, $\sin\alpha = \frac{2t}{1+t^2}$. RHS $= \begin{bmatrix} 1 & t \\ -t & 1 \end{bmatrix}\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} = \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix} = I + A$.

Q No.	Answer / Key Steps
19.	$A^2 = \begin{bmatrix} 19 & 4 & 8 \\ 1 & 12 & 8 \\ 14 & 6 & 15 \end{bmatrix}$, $A^3 = \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix}$; $A^3 - 23A - 40I = O$.
20.	(i) $\begin{bmatrix} 40 & 50 & 20 \\ 25 & 40 & 30 \\ 35 & 50 & 40 \end{bmatrix}$, $\begin{bmatrix} 25 \\ 100 \\ 50 \end{bmatrix}$. (ii) ₹7,000. (iii) P: ₹7,000; Q: ₹6,125; R: ₹7,875; total ₹21,000.